

test2 practice

 This is a preview of the published version of the quiz

Started: Apr 7 at 12:24pm

Quiz Instructions

It does not count toward your final grade.

https://www.youtube.com/watch?v=9O1J5_5UoqI  https://www.youtube.com/watch?v=9O1J5_5UoqI



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-> went through some problems for your

Topics : unit about potential. Including integrals. and DC circuits. capacitors too because its easy. So potential, DC circuit, capacitors. We didn't do capacitors yet but you can figure it out using the following equation.

A capacitor is made of 2 parallel plates. 2 conductors (we already covered that). We define the capacitance C is the amount of charge stored per unit volt. Unit is farad (F). A is the area of the plates/ d is the distance between them. U is the energy stored in the electric field E . E is between the plates. from + to -. k is the dielectric constant. If you place an insulator between the plates, it will boost the capacitance. k depends on the material. $k=1$ for air.

play with: <https://phet.colorado.edu/en/simulations/capacitor-lab>



Question 1 5 pts

An ideal parallel-plate capacitor having circular plates of diameter D that are a distance d apart stores energy U when it is connected across a fixed potential difference. If you want to triple the amount of energy stored in this capacitor by changing only the size of its plates, the diameter should be changed to

Check your equation sheet. (header) . Its basically Algebra. The area of of circle is proportional to the diameter squared ! Look at your equations. U is a function of capacitance C which is a function of area A



D/3



9 D



9D.

 $\sqrt{3} D$ 

3D.

**Question 2 5 pts**

A 200-W light bulb is connected across a 110-V dc power supply. What current will flow through this bulb?



0 A



0.90 A



0.36 A



1.8 A



0.60 A

**Question 3 5 pts**

A 1500-W heater is connected to a 120-V line for 2.0 hours. How much heat energy is produced?



1.5 kJ



0.18 MJ



11 MJ



3.0 kJ

**Question 4 5 pts**

How much does it cost to operate a 25-W soldering iron for 8.0 hours if energy costs 8.0¢/kWh?



\$1.50



16¢



25¢



1.6¢

**Question 5 5 pts**

What current is flowing in a resistor if 4.0×10^{16} electrons pass a point in the resistor in 0.50 s? ($e = 1.60 \times 10^{-19}$ C)



0.013 A



0.31 A



78 A



6.3 A

**Question 6 5 pts**

Two large parallel plates are separated by 1.0 mm of air. If the potential difference between them is 3.0 V, what is the magnitude of their surface charge densities? ($\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2$)

 $5.3 \times 10^{-8} \text{ C/m}^2$  $3.3 \times 10^{-4} \text{ C/m}^2$  $2.7 \times 10^{-8} \text{ C/m}^2$  $1.6 \times 10^{-4} \text{ C/m}^2$ **Question 7 5 pts**

The potential difference between the plates of an ideal air-filled parallel-plate capacitor with a plate separation of 6.0 cm is 60 V. What is the strength of the electric field between the plates of this capacitor?



1000 V/m



2000 V/m



60 V/m



500 V/m



3600 V/m

**Question 8 5 pts**

How much energy is necessary to place three $+2.0\text{-}\mu\text{C}$ point charges at the vertices of an equilateral triangle of side 2.0 cm if they started out extremely far away? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

Use the equation for electrostatic energy U. If its negative, its binding.



5.4 J



7.6 J



6.7 J



4.5 J

**Question 9 5 pts**

A proton with a speed of $2.0 \times 10^5 \text{ m/s}$ accelerates through a potential difference and thereby increases its speed to $4.0 \times 10^5 \text{ m/s}$. Through what magnitude potential difference did the proton accelerate? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)



840 V



1000 V



100 V



210 V



630 V

**Question 10 5 pts**

What is the speed of a proton that has been accelerated from rest through a potential difference of 4.0 kV?

 $1.1 \cdot 10^6 \text{ m/s}$  $8.8 \cdot 10^5 \text{ m/s}$



$$6.2 \cdot 10^5 \text{ m/s}$$

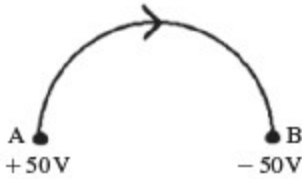


$$9.8 \cdot 10^5 \text{ m/s}$$



Question 11 5 pts

How much work must we do on an electron to move it from point A, which is at a potential of +50V, to point B, which is at a potential of -50 V, along the semicircular path shown in the figure? Assume the system is isolated from outside forces. ($e = 1.60 \times 10^{-19} \text{ C}$)



$$-1.60 \times 10^{-17} \text{ J}$$



$$-1.6 \text{ J}$$



$$1.60 \times 10^{-17} \text{ J}$$



$$1.6 \text{ J}$$



This cannot be determined because we do not know the distance traveled.



Question 12 5 pts

A 0.25-mF capacitor is connected to a 400-V battery. What potential energy is stored in the capacitor?



$$1.2 \cdot 10^{-12} \text{ J}$$



$$1.0 \cdot 10^{-4} \text{ J}$$



$$0.04 \text{ J}$$



$$0.020 \text{ J}$$



Question 13 5 pts

A rod is 4.0 m long and has a square cross-section that is 1.5 cm on each side. An ohmmeter measures $0.040 \, \Omega$ across its ends. What is the resistivity of the material from which this rod is made?

0.023 $\Omega \cdot \text{m}$  $1.5 \times 10^{-4} \Omega \cdot \text{m}$ 0.015 $\Omega \cdot \text{m}$  $2.3 \times 10^{-6} \Omega \cdot \text{m}$ **Question 14 5 pts**

A parallel-plate capacitor is connected to a battery and becomes fully charged. The capacitor is then disconnected, and the separation between the plates is increased in such a way that no charge leaks off. As the plates are being separated, the energy stored in this capacitor



does not change.



become zero.



decreases.



increases.

**Question 15 5 pts**

A proton ($m = 1.7 \times 10^{-27} \text{ kg}$, $q = +1.6 \times 10^{-19} \text{ C}$) starts from rest at point A and has a speed of 40 km/s at point B. Only electric forces act on it during this motion. Determine the electric potential difference $V_B - V_A$.



-4.8V



-8.5 V



4.8V



8.5V

**Question 16 5 pts**

Doubling the capacitance of a capacitor that is holding a constant charge causes the energy stored in that capacitor to



decrease to one-half.



quadruple.



decrease to one-fourth.



double.

**Question 17 5 pts**

A light bulb operating at a dc voltage of 120 V has a resistance of 200 Ω . How much power is dissipated in this bulb?



14 mW



7.2 W



100 W



72 W



60 W

**Question 18 5 pts**

A point charge of +3.0 mC is located at the origin of a coordinate system and a second point charge of -6.0 mC is at $x = 1.00$ m. What is the electric potential at the $x = 0.50$ m point? ($k_e = 8.99 \times 10^9$

N $\times$ m²/C²)

 -11×10^4 V -5.4×10^4 V 11×10^4 V 16×10^4 V**Question 19 5 pts**

A solid conducting sphere (radius = 5.0 cm) has a charge of 0.25 nC distributed uniformly on its surface. If point A is located at the center of the sphere and point B is 15 cm from the center, what is the magnitude of the electric potential difference between these two points?

hint: the potential on the surface is the same inside the sphere



23V

☐

30V

☐

45V

☐

23V

☐

Question 20 5 pts

Charge of uniform linear density 6.0 nC/m is distributed along the x axis from $x = 0$ to $x = +3$ m. Which of the following integrals is correct for the electric potential (relative to zero at infinity) at the point $y = +4$ m on the y axis?

☐ $\int_0^3 \frac{108 dx}{(x^2 + 16)^{1/2}}$

☐ $\int_0^3 \frac{54 dx}{(x^2 + 16)^{1/2}}$

☐ $\int_0^3 \frac{18 dx}{(x^2 + 16)^{1/2}}$

☐ $\int_0^3 \frac{54 dx}{x^2 + 16}$

A. B, C, D

☐

D

☐

C

☐

A

☐

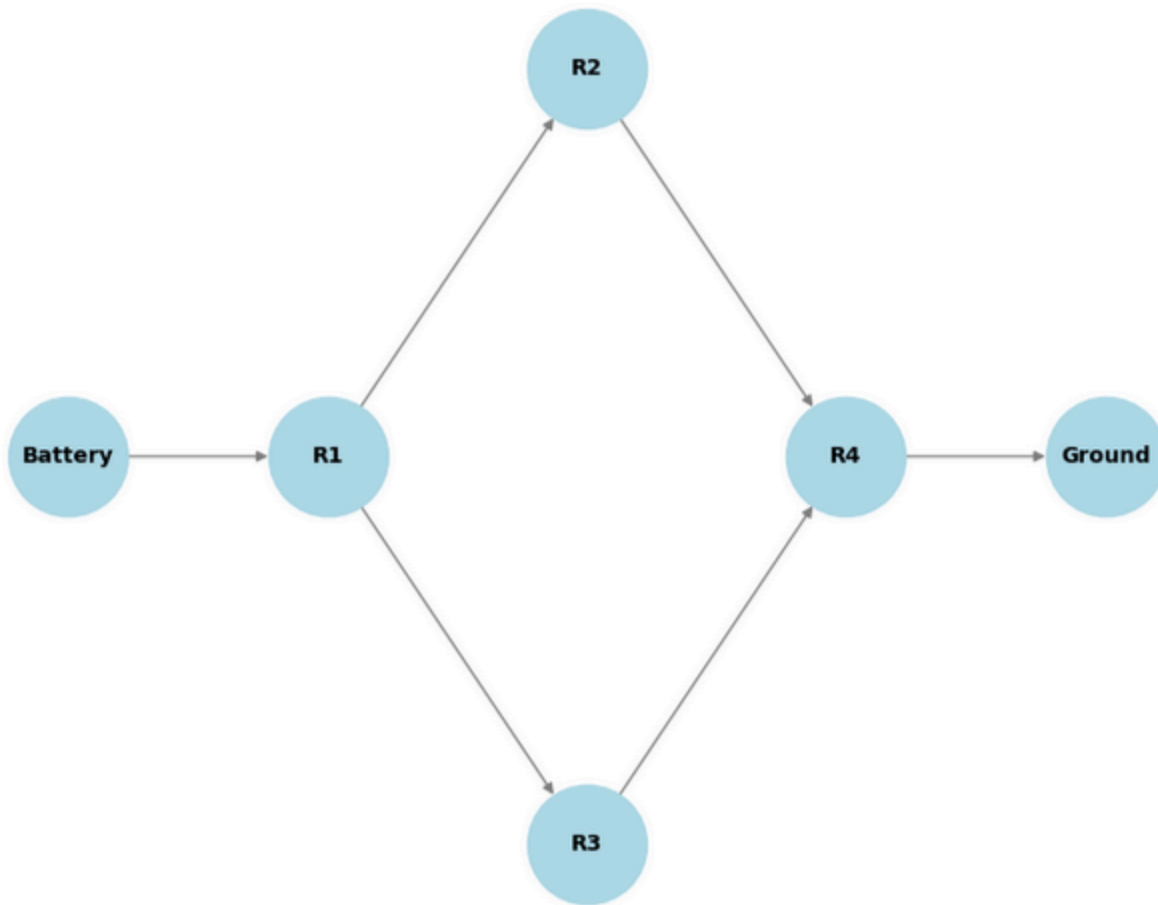
B

☐

Question 21 1 pts



Circuit Diagram: Parallel Branches in Series with a Resistor



hint: (hide first) - test yourself

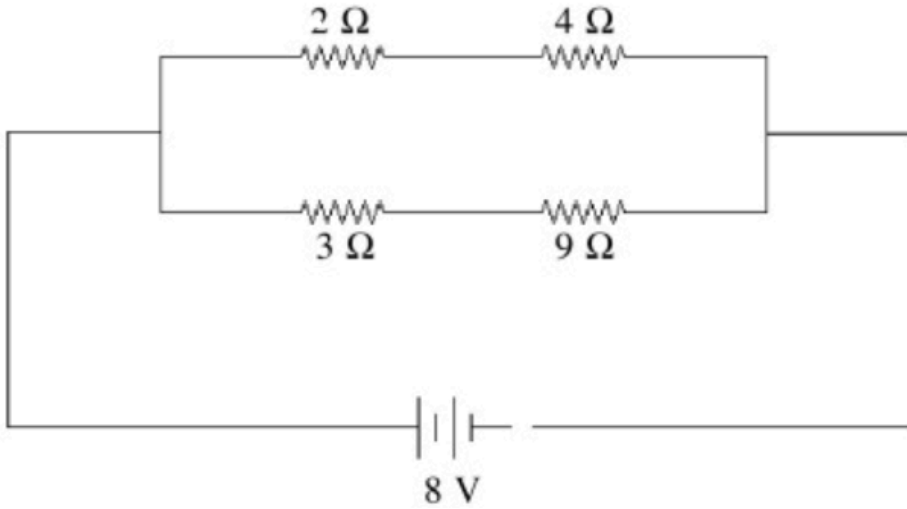
battery R2 and R3 and in parallel. and these branches are in series with R1 and R4.

- ☐ about 70 ohms
- ☐ about 60 ohms
- ☐ about 50 ohms
- ☐ about 80 ohms



Question 22 1 pts

9) Four resistors are connected across an 8-V DC battery as shown in the figure. The current through the $2\text{-}\Omega$ resistor is closest to



- ☐ 0.7 A
- ☐ 2 A
- ☐ 2.7 A
- ☐ 1,333 A

Not saved

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